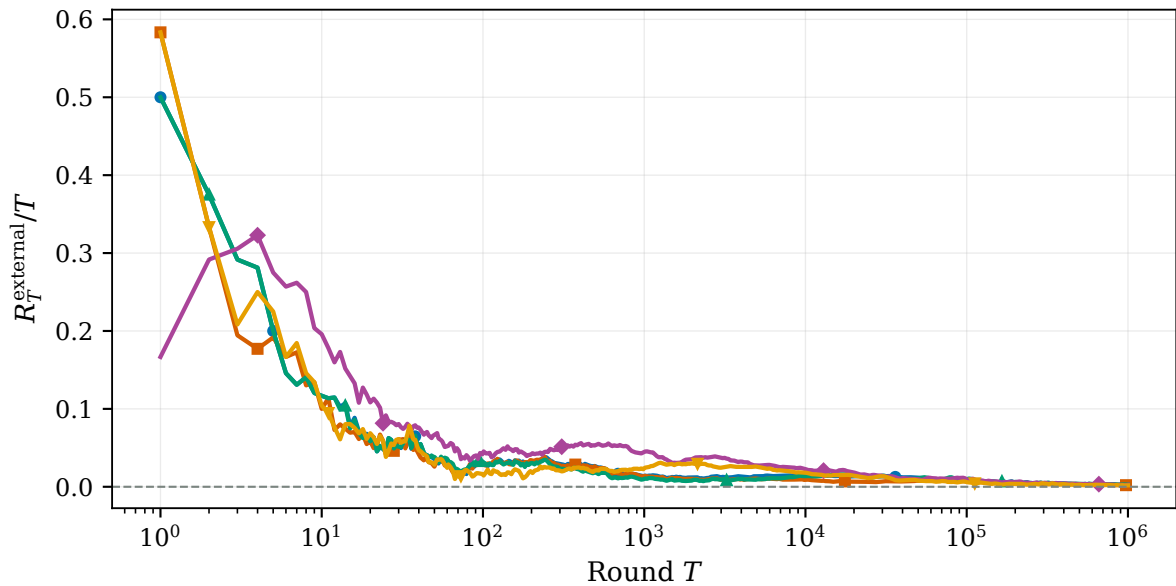


Experiment Information

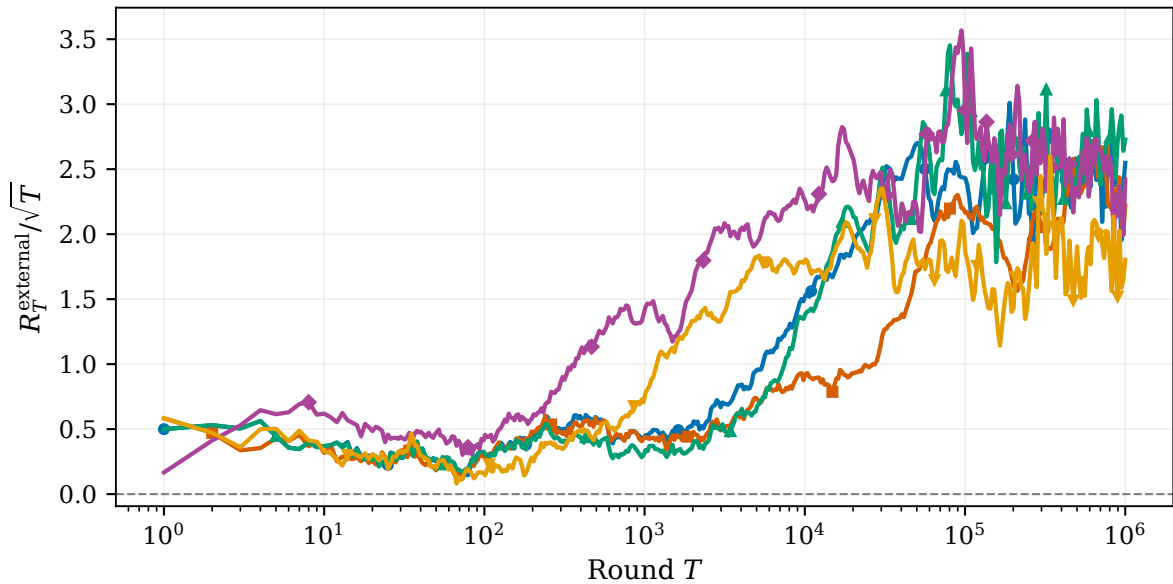
Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 0.00254787, SD = 0.00137973, SE = 0.000398293
BM-EXP3 vs BM-EXP3:	mean = 0.00221958, SD = 0.00128487, SE = 0.000370911
EXP3-IX vs EXP3-IX:	mean = 0.00272675, SD = 0.000962414, SE = 0.000277825
Ito-Tsallis vs Ito-Tsallis:	mean = 0.00242054, SD = 0.000822685, SE = 0.000237489
LCE-IX vs LCE-IX:	mean = 0.00180312, SD = 0.000521786, SE = 0.000150627



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|--------------------|----------------------------|
| EXP3 vs EXP3 | Ito-Tsallis vs Ito-Tsallis |
| BM-EXP3 vs BM-EXP3 | LCE-IX vs LCE-IX |
| EXP3-IX vs EXP3-IX | |

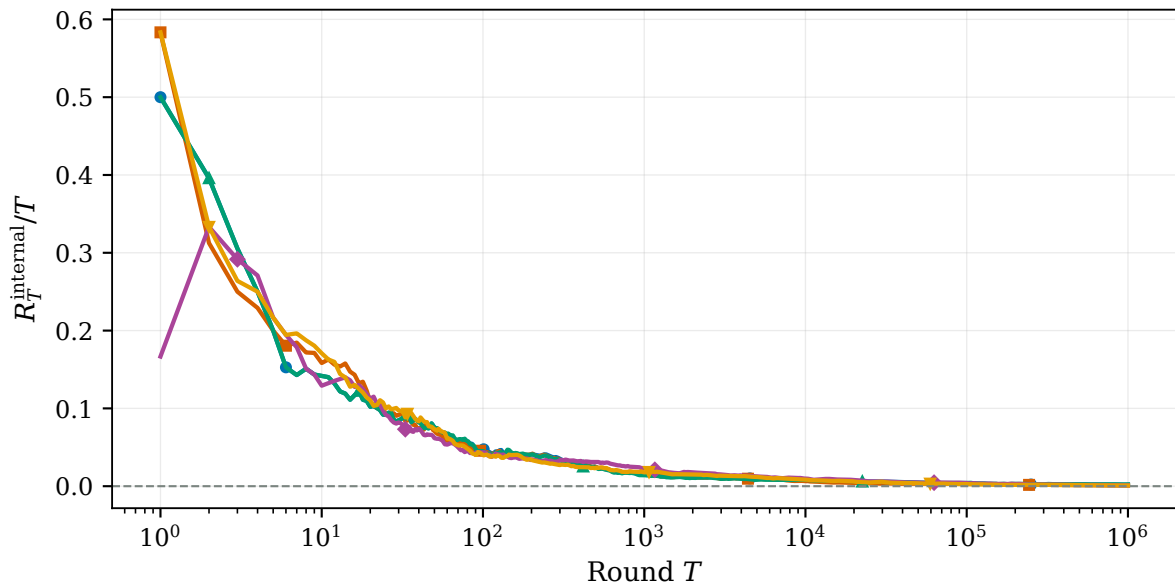
Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 2.54787, SD = 1.37973, SE = 0.398293
BM-EXP3 vs BM-EXP3:	mean = 2.21958, SD = 1.28487, SE = 0.370911
EXP3-IX vs EXP3-IX:	mean = 2.72675, SD = 0.962414, SE = 0.277825
Ito-Tsallis vs Ito-Tsallis:	mean = 2.42054, SD = 0.822685, SE = 0.237489
LCE-IX vs LCE-IX:	mean = 1.80312, SD = 0.521786, SE = 0.150627



Experiment Information

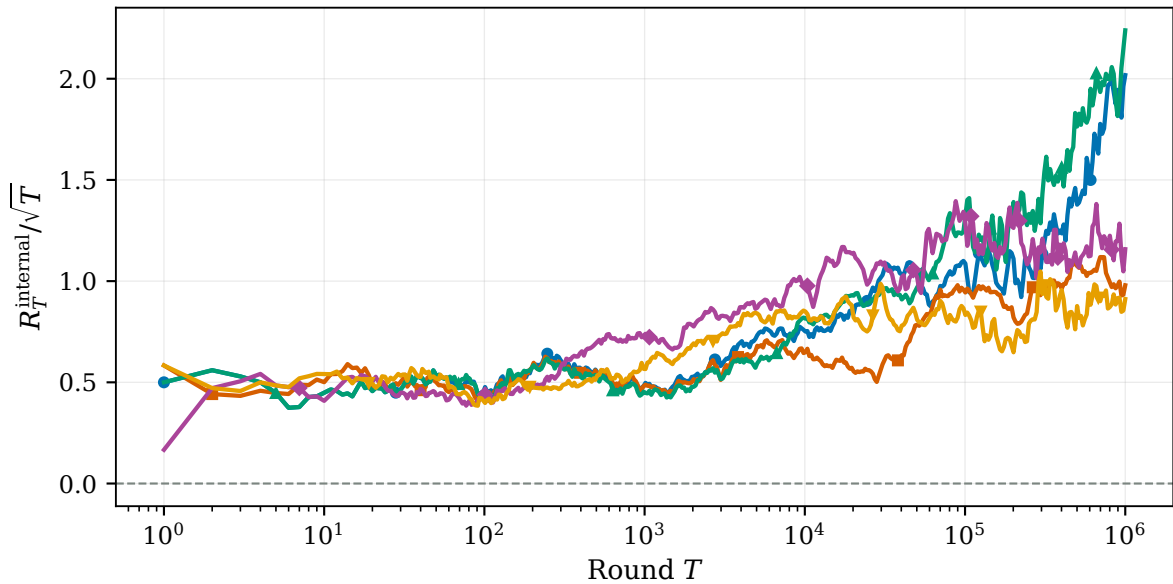
Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 0.00201763, SD = 0.000852046, SE = 0.000245964
BM-EXP3 vs BM-EXP3:	mean = 0.000979542, SD = 0.000315161, SE = 9.0979e-05
EXP3-IX vs EXP3-IX:	mean = 0.00223908, SD = 0.000743161, SE = 0.000214532
Ito-Tsallis vs Ito-Tsallis:	mean = 0.00115775, SD = 0.000274926, SE = 7.93643e-05
LCE-IX vs LCE-IX:	mean = 0.000911417, SD = 0.000225516, SE = 6.51008e-05



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|--------------------|----------------------------|
| EXP3 vs EXP3 | Ito-Tsallis vs Ito-Tsallis |
| BM-EXP3 vs BM-EXP3 | LCE-IX vs LCE-IX |
| EXP3-IX vs EXP3-IX | |

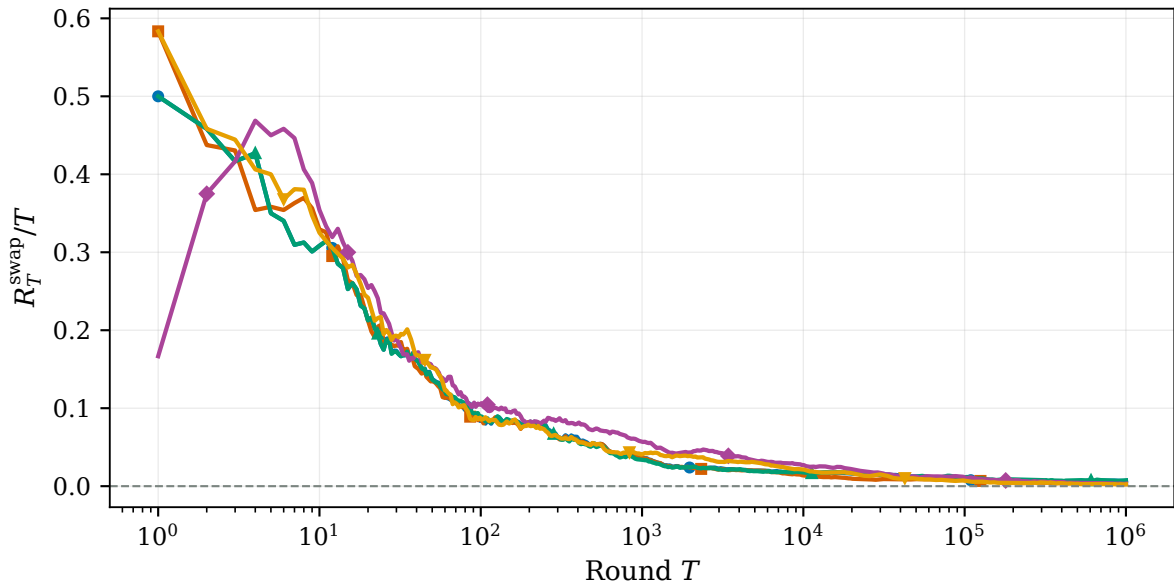
Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 2.01763, SD = 0.852046, SE = 0.245964
BM-EXP3 vs BM-EXP3:	mean = 0.979542, SD = 0.315161, SE = 0.090979
EXP3-IX vs EXP3-IX:	mean = 2.23908, SD = 0.743161, SE = 0.214532
Ito-Tsallis vs Ito-Tsallis:	mean = 1.15775, SD = 0.274926, SE = 0.0793643
LCE-IX vs LCE-IX:	mean = 0.911417, SD = 0.225516, SE = 0.0651008



Experiment Information

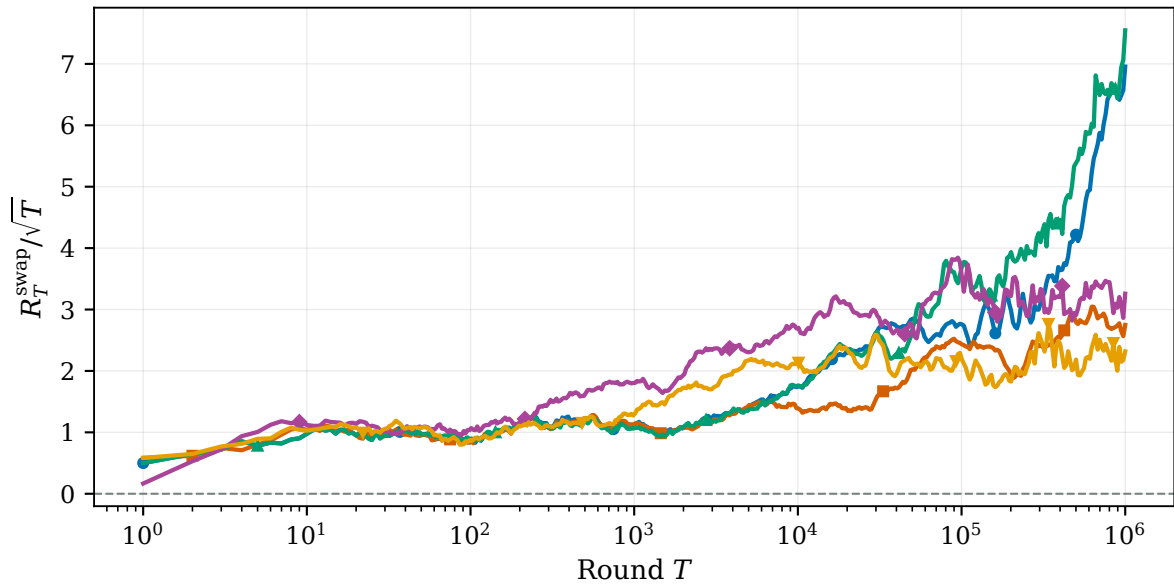
Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 0.00695588, SD = 0.00335592, SE = 0.000968771
BM-EXP3 vs BM-EXP3:	mean = 0.00274821, SD = 0.00122064, SE = 0.000352369
EXP3-IX vs EXP3-IX:	mean = 0.00754608, SD = 0.00290704, SE = 0.000839189
Ito-Tsallis vs Ito-Tsallis:	mean = 0.00325787, SD = 0.000808102, SE = 0.000233279
LCE-IX vs LCE-IX:	mean = 0.00231804, SD = 0.000469859, SE = 0.000135637



- EXP3 vs EXP3
- BM-EXP3 vs BM-EXP3
- EXP3-IX vs EXP3-IX
- Ito-Tsallis vs Ito-Tsallis
- LCE-IX vs LCE-IX

Experiment Information

Figure:	Algorithm comparison
Game:	Rock–Paper–Scissors–Lizard–Spock
Player:	0
Feedback:	Bandit feedback
Algorithms:	EXP3 vs EXP3, BM-EXP3 vs BM-EXP3, EXP3-IX vs EXP3-IX, Ito-Tsallis vs Ito-Tsallis, LCE-IX vs LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
EXP3 vs EXP3:	mean = 6.95588, SD = 3.35592, SE = 0.968771
BM-EXP3 vs BM-EXP3:	mean = 2.74821, SD = 1.22064, SE = 0.352369
EXP3-IX vs EXP3-IX:	mean = 7.54608, SD = 2.90704, SE = 0.839189
Ito-Tsallis vs Ito-Tsallis:	mean = 3.25787, SD = 0.808102, SE = 0.233279
LCE-IX vs LCE-IX:	mean = 2.31804, SD = 0.469859, SE = 0.135637



- EXP3 vs EXP3
- BM-EXP3 vs BM-EXP3
- EXP3-IX vs EXP3-IX
- Ito-Tsallis vs Ito-Tsallis
- LCE-IX vs LCE-IX